

Conclusion

The Human Development Index (HDI) Prediction System demonstrates how machine learning can be used to estimate a country's level of human development using important socio-economic indicators such as life expectancy, mean years of schooling, gross national income (GNI) per capita, and internet users. By applying data preprocessing techniques, feature selection, model training, and evaluation, the developed Linear Regression model is capable of generating reliable HDI predictions that closely match the actual values. The integration of the trained model into a Flask-based web application provides a simple and user-friendly interface that allows users to enter input values and instantly obtain the predicted HDI score along with its corresponding development category. This makes the system practical for educational purposes, research, and decision support. The project also highlights the importance of measuring development using a comprehensive view of well-being rather than relying solely on economic indicators. Human development is influenced by multiple interconnected dimensions, including health, education, income, and access to opportunities. A multidimensional assessment provides a more meaningful understanding of the quality of life and supports informed policy decisions aimed at sustainable and inclusive development. Overall, the proposed HDI Prediction System successfully combines machine learning and web technologies to create an efficient, accurate, and accessible prediction platform. The project demonstrates the potential of artificial intelligence in supporting data-driven decision-making and can be further enhanced by incorporating additional development indicators, larger datasets, and advanced machine learning algorithms to improve prediction accuracy and real-world applicability.